

# Analysis group meeting

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09/03/24

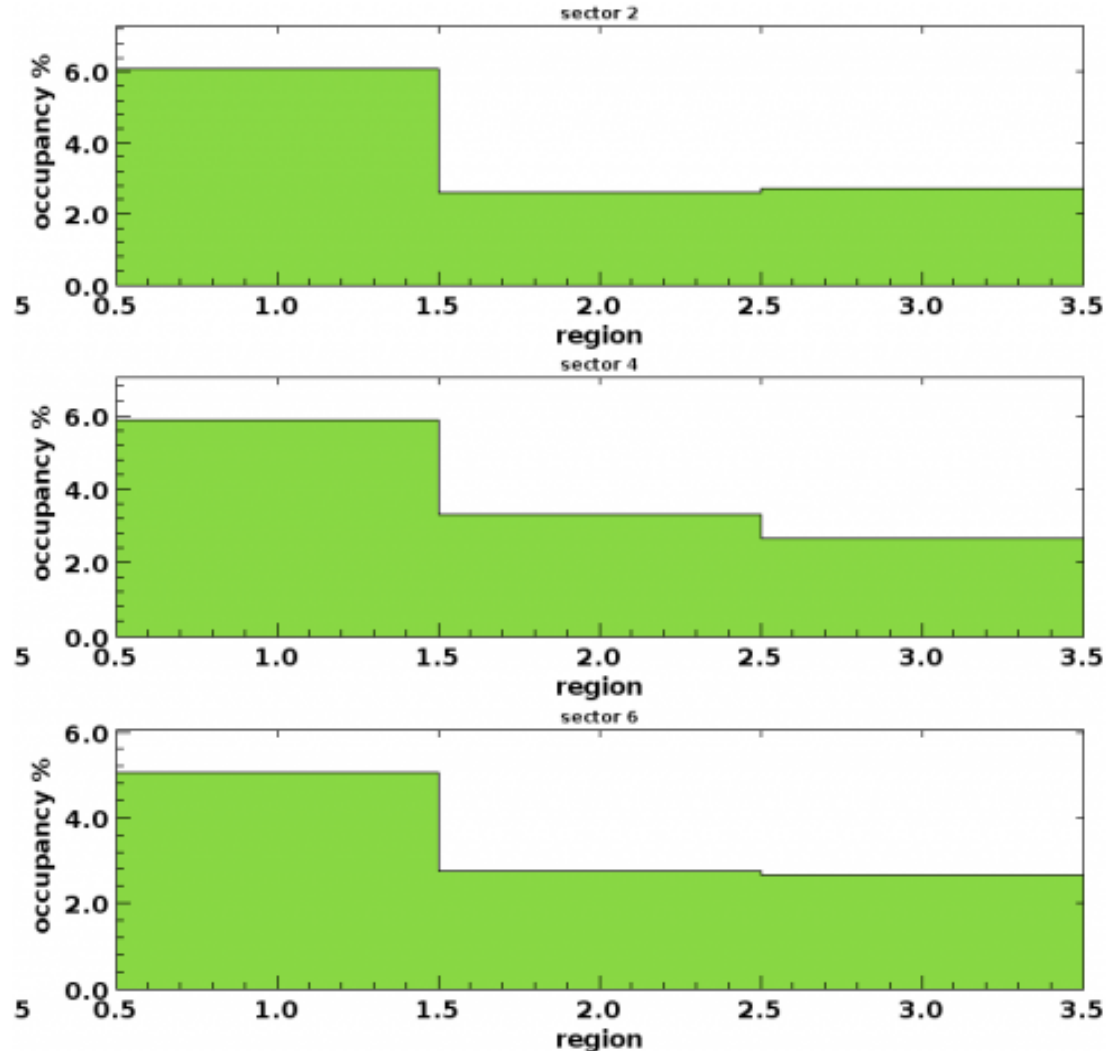


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# Overview

- ALERT occupancy study
  - Florian point out a problem with RG-F occupancy
  - Simulate RG-F occupancy
  - Simulate ALERT occupancy with a shift of the central detector and the target
- RG-D studies
  - Extract nuclear transparency
  - Compare central and forward pion +/-
  - $V_z$  study for the lower cut.
- Other
  - Get the RWI training
  - You can find these presentation here :  
[https://github.com/mathieuouillon/Presentation/tree/main/Lamiaa\\_meeting](https://github.com/mathieuouillon/Presentation/tree/main/Lamiaa_meeting)

# RG-F real data occupancy



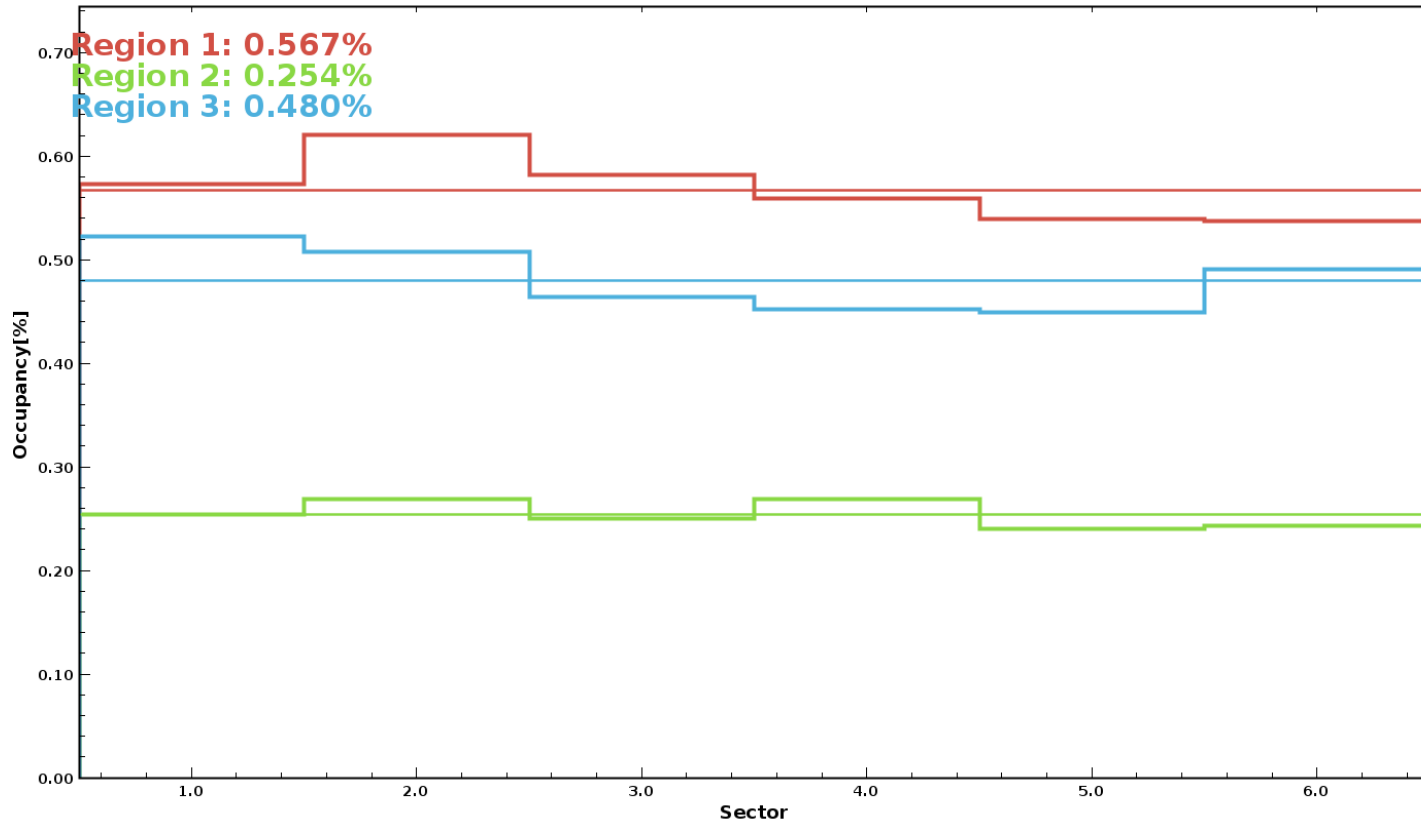
DC rate appears reasonable in the simulation. However, they were much higher with Bonus

For RG-F/BONuS12 experiment, they use a 250 nA current on a similar setup. They get a occupancy for region 1 around 6%. We will use a higher current on a smaller target (45- >55cm) .

But our occupancy on the simulation is 3.5% in region 1.

The occupancy from the online monitoring histograms use the trigger electron. Which is not the case in the simulation.

# Try the RG-F simulation to compare



RG-F FTOff, 250 nA -> 387,796 electrons.

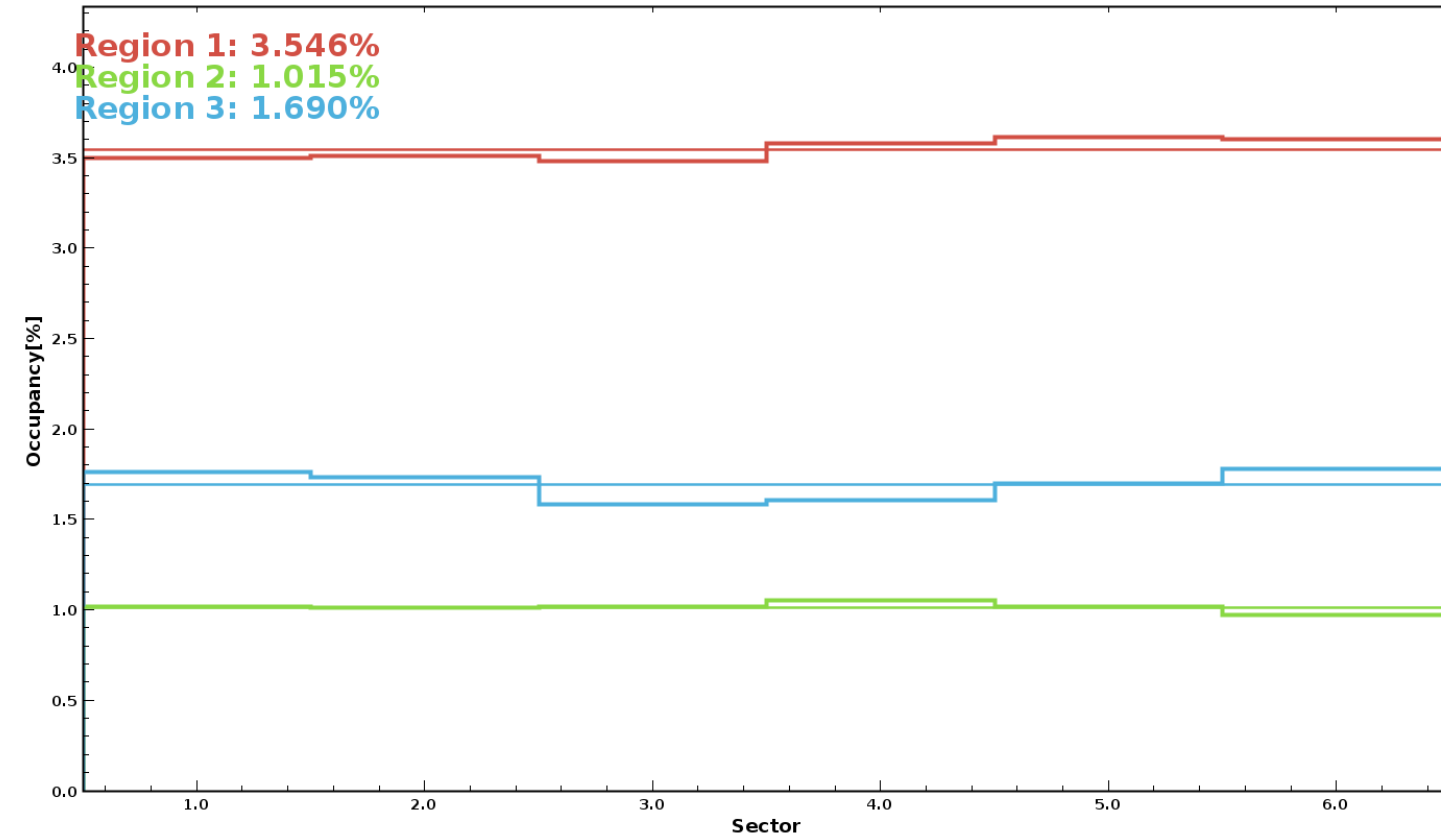
Much lower occupancy 0.567% in region 1.

Could come from a non complete simulation.

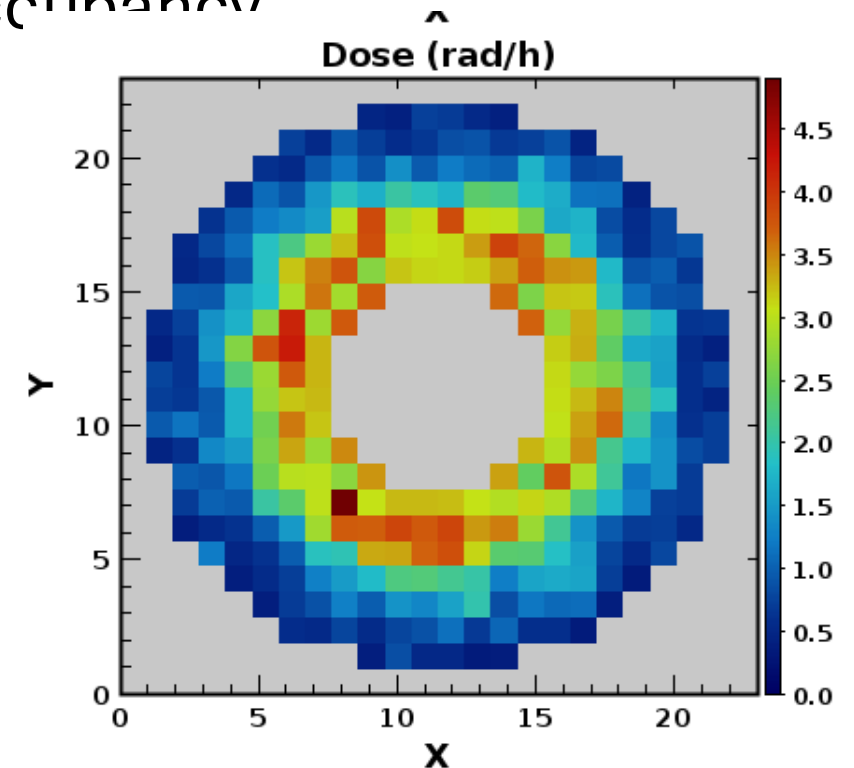
They have only one windows for the target for example.

Need to understand that. I could have made a mistake on the simulation.

# Add a 5cm shift on CD and target



I moved the target and all the central detectors (ALERT/CND/CTOF) and the helium bag 5cm upstream (nearer to the beam) to assess whether such a shift would significantly impact the occupancy.



Bit higher occupancy/dose in DC of FT. Compare to the nominal setup. Could be statistic (20k to 50k)

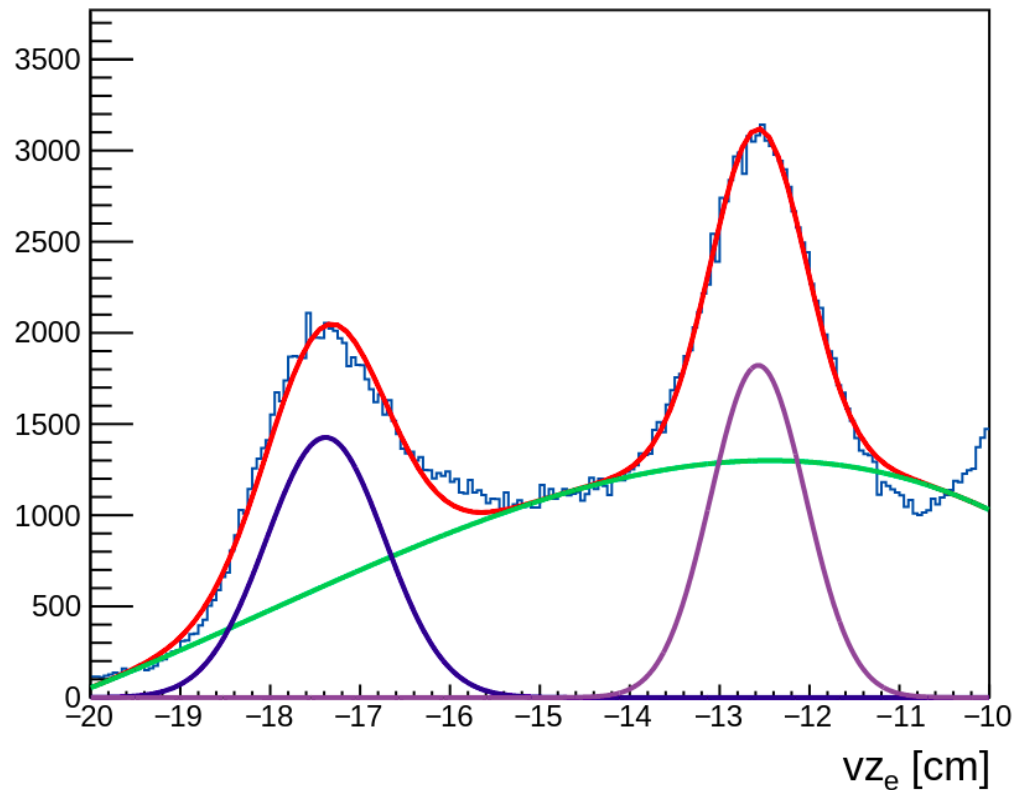
# RG-D study vz\_electron cut

Bspot carbon runs, 10% of the all stats

Clearly I have a problem since with the all stat, it doesn't work.

Use 2 gaussian and a polynom 3rd order

CxC

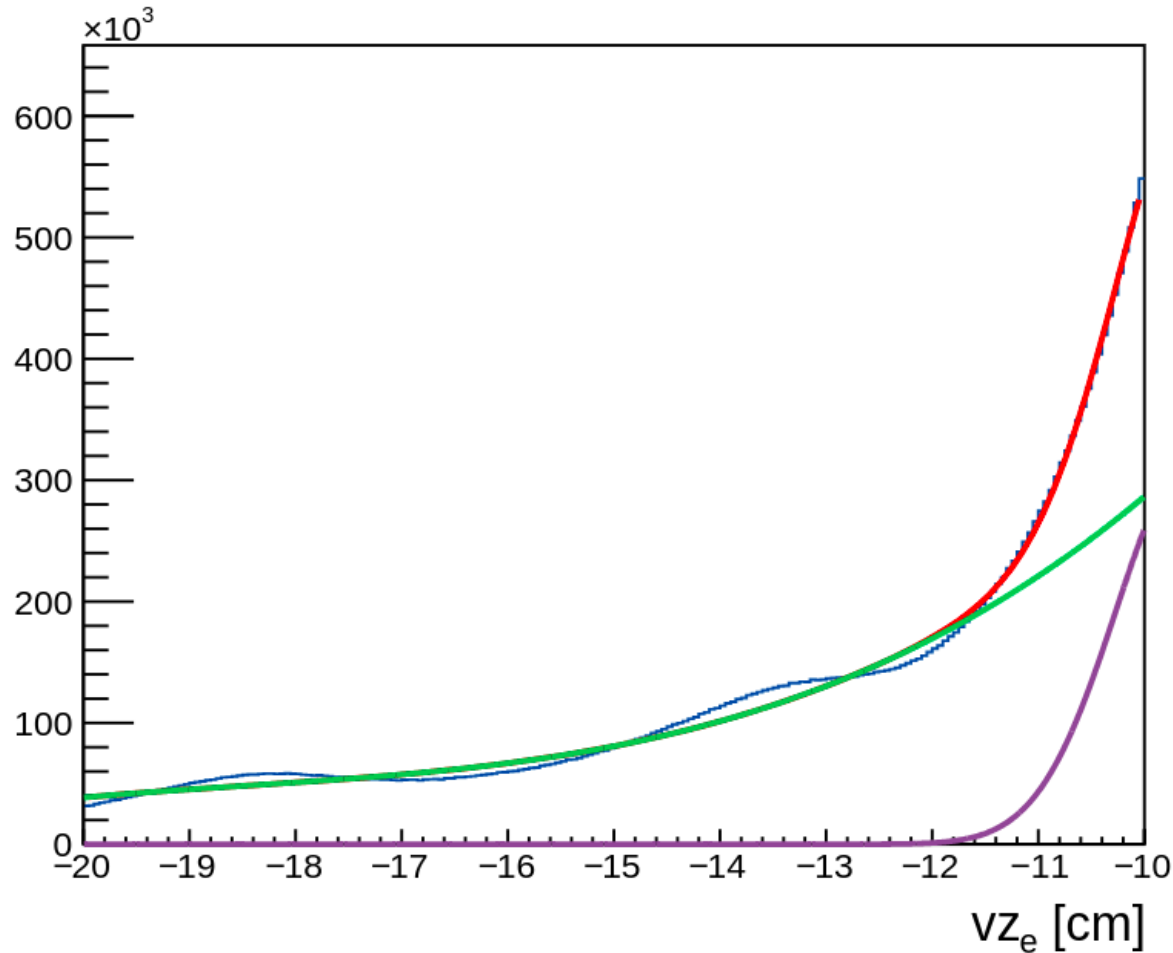


|                       |                     |
|-----------------------|---------------------|
| Entries               | 7087941             |
| Mean                  | -14.25              |
| Std Dev               | 2.481               |
| $\chi^2 / \text{ndf}$ | 1338 / 190          |
| Prob                  | 1.181e-171          |
| p0                    | -17.38 $\pm$ 0.01   |
| p1                    | 0.6613 $\pm$ 0.0050 |
| p2                    | 2364 $\pm$ 18.5     |
| p3                    | -12.57 $\pm$ 0.00   |
| p4                    | 0.5272 $\pm$ 0.0043 |
| p5                    | 2408 $\pm$ 21.5     |
| p6                    | -9369 $\pm$ 14.2    |
| p7                    | -2080 $\pm$ 1.4     |
| p8                    | -127.6 $\pm$ 0.1    |
| p9                    | -2.356 $\pm$ 0.003  |

# RG-D studies vz\_electron cut

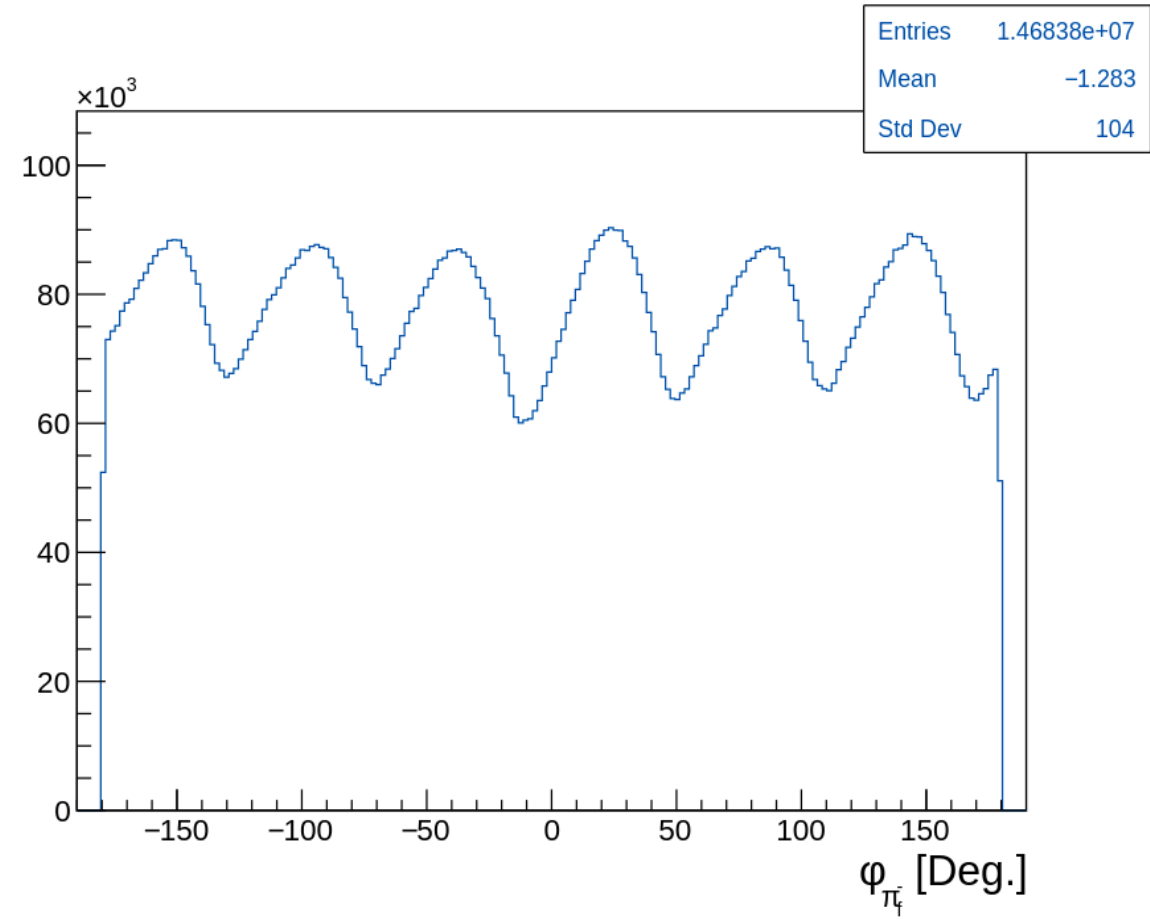
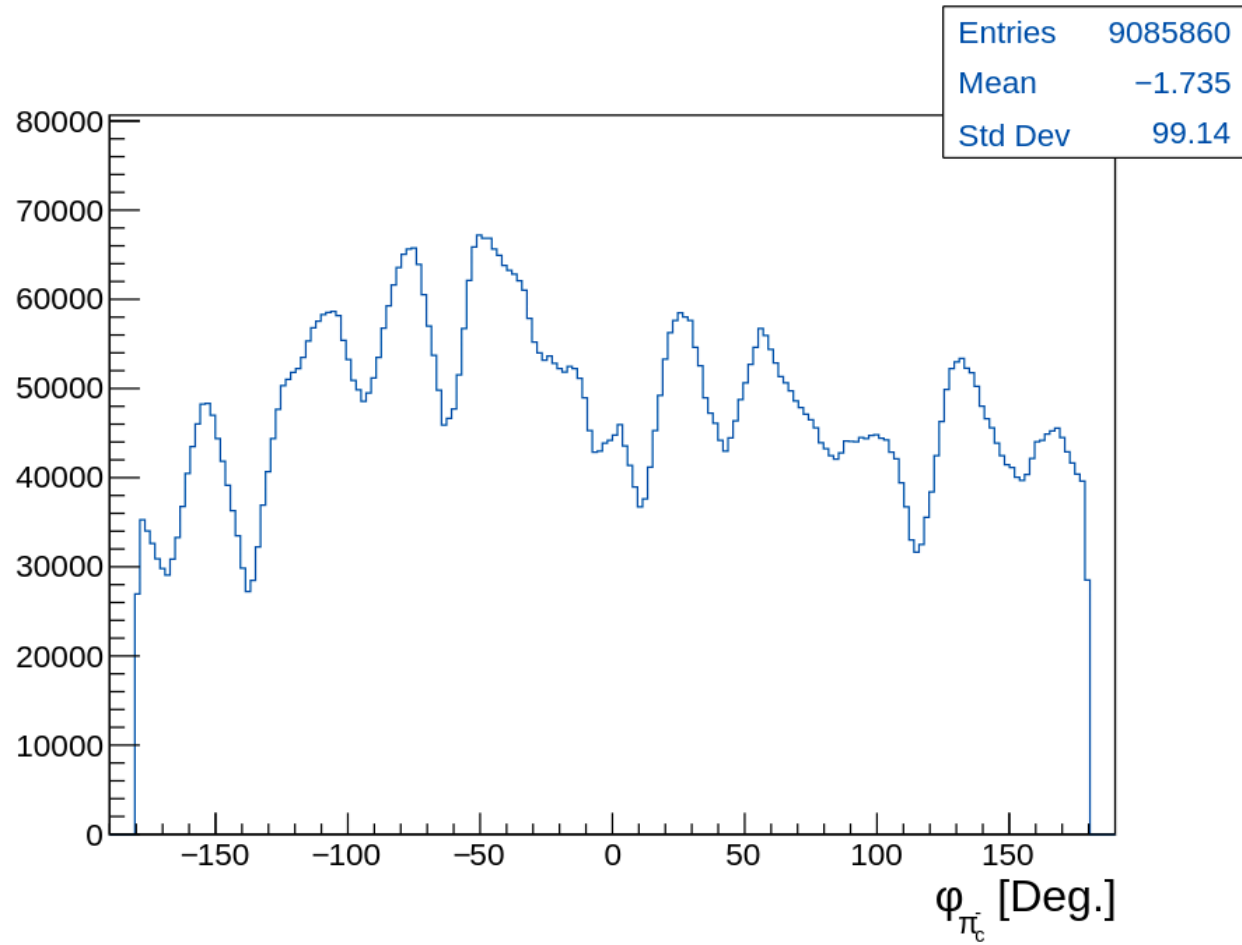
With all the stats, the fit fail.

CxC



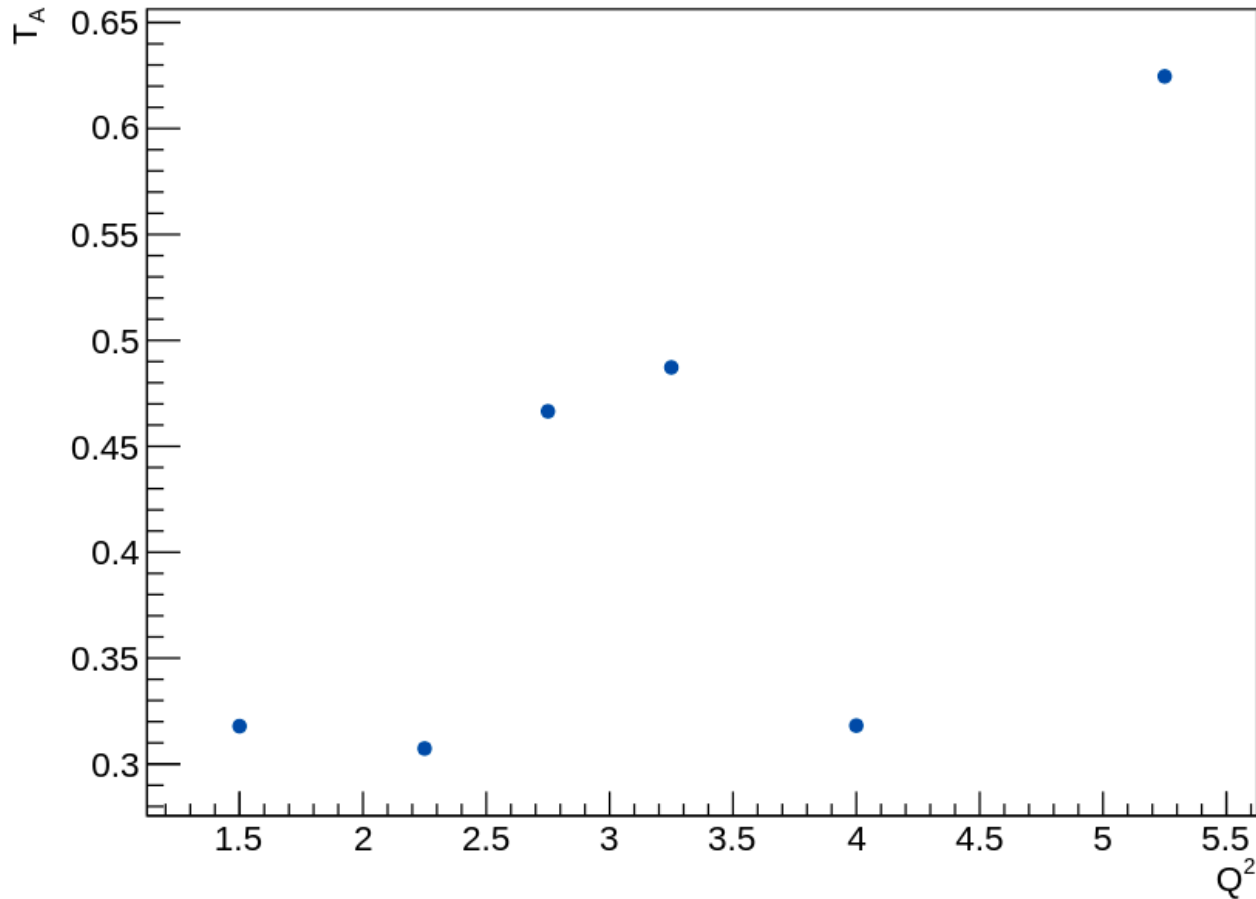
|                       |                           |
|-----------------------|---------------------------|
| Entries               | 1.816433e+08              |
| Mean                  | -12.99                    |
| Std Dev               | 2.697                     |
| $\chi^2 / \text{ndf}$ | 1.058e+05 / 190           |
| Prob                  | 0                         |
| p0                    | 56.07 $\pm$ 3.36          |
| p1                    | 13.53 $\pm$ 0.01          |
| p2                    | -1921 $\pm$ 1477.4        |
| p3                    | -9.527 $\pm$ 0.009        |
| p4                    | -0.7373 $\pm$ 0.0035      |
| p5                    | 5.896e+05 $\pm$ 7.272e+03 |
| p6                    | 2.131e+06 $\pm$ 7.936e+03 |
| p7                    | 3.271e+05 $\pm$ 1.509e+03 |
| p8                    | 1.741e+04 $\pm$ 9.451e+01 |
| p9                    | 314.4 $\pm$ 2.0           |

# Central vs forward pions





# Nuclear transparency



- Use bspot runs CxC and LD2.
- Trigger electron status  $< 0$  and  $-3 < \chi^2_{\text{pid}} < 3$  |  $-12 < v_z < 0$  cm
- Pion+/-  $-3 < \chi^2_{\text{pid}} < 3$
- Fit the invariant mass distribution for different  $Q^2$  bins.
- Extract the yield by integrating.
- Compute the nuclear transparency.

# To-do

- ALERT Occupancy :
  - Fix the RG-F beamline to match the online monitoring histograms.
  - Raphael ask me to give a presentation for the hall B staff meeting next monday on simulation and occupancy study
- RG-D study :
  - Fix the fit for the the vz study when I use the all stats.
  - Finish the nuclear transparency (error bars)
  - Setup a meeting with Matt to cross check the number of  $\rho_0$  and the nuclear transparency.
- Cooking chef :
  - Check runs use for each target.